

4.8 Putting It All Together

Lesson Introduction

 Benchmarks	MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure. MA.912.GR.2.6 Apply rigid transformations to map one figure onto another to justify that the two figures are congruent.		 Learning Targets	- I can determine a sequence of transformations that map one figure onto another congruent figure. - I can match the written and algebraic descriptions of a sequence of transformations given the preimage and image of a figure.
 Benchmark Coherence	Building On MA.8.GR.2.1 Given a preimage and image generated by a single transformation, identify the transformation that describes the relationship.		Working Toward N/A	
 Essential Question	How can I apply what I have learned in this unit to determine a sequence of transformations that map one figure onto another congruent figure?			
 Lesson Narrative	This lesson serves as a culminating experience for this unit. Students will apply all that they have learned throughout the unit to complete a card sort where they will match a coordinate grid that shows the preimage and image of a triangle and match it with its appropriate written description and algebraic description.			
 Component Summary	4.8.1 Warm-Up (~5 min.)	Notice and Wonder of a diagram activity (<i>SE p. 218</i>)	4.8.2 Activity (30-35 min.)	Card Sort activity with matching a coordinate grid with the correct written description and algebraic description (<i>SE p. 218</i>)
	4.8.3 Wrap-Up (~5 min.)	Self-reflection activity on current learning for the learning targets for this lesson (<i>SE p. 219</i>)		
 Resources	Glossary		Additional Preparation	
	Key Terms: N/A Additional Terms: N/A		Go through the Card Sort activity on your own prior to having students do it. This will help you look for potential errors or mismatches that may occur. By doing this, you can quickly observe where a group may be going wrong while facilitating.	
	Materials			
	Card sort activity cards			

 Teacher Tips	Common Misconceptions - Students may misinterpret the preimage vs the image in the coordinate grid cards. - Students may mistake their signs when looking at the algebraic description cards. - Students may misread the written description cards in this activity.	Things to Consider This lesson is primarily the 4.8.2 Activity Card Sort. It will require physical materials (the cards for the activity) to be prepared for students prior to the lesson. Consider having this done before the lesson to allow more time for the actual activity itself.
	Literacy and Language Routines Notice and Wonder The 4.8.1 Warm-Up leverages the Notice and Wonder protocol. Encourage low-floor answers like the difference in the type of grid while also prompting students to extend their answers for high-ceiling statements that include learning from the unit like rigid motions, isometry, or congruence.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.2.1: Multiple Representations In the 4.8.2 Activity, students will be completing a Card Sort where they match a transformation provided in the coordinate grid (preimage and image), written description, and algebraic description.
	 Differentiate Instruction Consider the placement of students in their small groups based on successes observed with each of the individual transformations in the previous lessons of this unit. This will allow students an entry point of using their individual strengths during the Card Sort activity. Also consider eliminating some of the cards for a shorter activity. Suggested cards for eliminating are card 5, card 6, card 9, and card 10. Consider using the Card Sort with a group of students who need further support in small group instruction.	
Lesson Notes:		

4.8.1 Warm-Up: Notice and Wonder

Component Overview: Students will engage in a Notice and Wonder activity, examining a quadrilateral that has been reflected and translated.



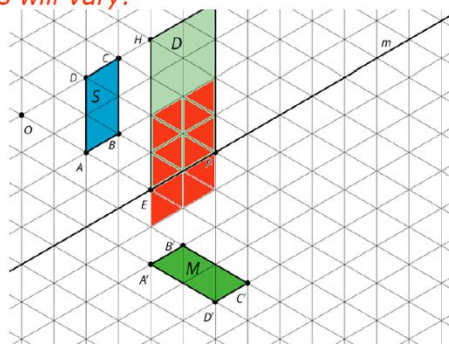
4.8.1 Warm-Up: Notice and Wonder

1. What are three things you notice about the diagram?

Answers will vary. I noticed that the three quadrilaterals are different sizes, they are different colors and two of them are the same size. I noticed that image D has 16 triangles while M and S have 4, D and M are shades of green while S is blue, M uses prime notation and S and D do not.

2. Reflect quadrilateral $EFGH$ across line m , then translate it up 3 units. Name the resulting quadrilateral $E'F'G'H'$.

Answers will vary.



3. Check your location of quadrilateral $E'F'G'H'$ with your partner. If your locations differ, explain why both could be correct. If your locations are the same, find another possible interpretation of up 3 units and draw that quadrilateral.

Answers will vary. My partner and I had the same answer but "up three" could have gone to the right instead. My partner reflected the same but went to the right instead of going vertically.



The Notice and Wonder intentionally provides opportunity for students to explore isometric graphing and its usefulness. Prompt students to notice visual differences while extending some student responses to include the usefulness of isometric paper when showing congruence. Encourage productive struggle and partner talk.

4.8.2 Activity: Card Sort

Component Overview: Students will work collaboratively with a partner or small groups to match graphed transformation, written description, and algebraic description cards that represent the same sequence of transformations. For this activity, consider reviewing the instructions and logistics of the activity prior to launching. Leave time in this activity for a review of the correct answer and sharing of the processes involved.



4.8.2 Activity: Card Sort

Match each coordinate grid with the preimage of ΔTLW and image ΔHPY where $\Delta TLW \cong \Delta HPY$, with the written and algebraic descriptions that map ΔTLW onto ΔHPY . Use the letters in the blue, green, and orange boxes to record your matches in the table.

Coordinate Grid	Written Description	Algebraic Description
B	<i>W</i>	<i>AA</i>
D	<i>N</i>	<i>BB</i>
F	<i>J</i>	<i>EE</i>
G	<i>K</i>	<i>GG, HH, or YY</i>
H	<i>S</i>	<i>CC</i>
M	<i>V</i>	<i>ZZ</i>
T	<i>L</i>	<i>GG, HH, or YY</i>
W	<i>C</i>	<i>FF</i>
X	<i>Q</i>	<i>GG, HH, or YY</i>
Y	<i>P</i>	<i>DD</i>



Consider modeling one example for instructions. For example, show that a “match” is a graphed transformation, a written description card, and an algebraic description card where all three represent the same sequence of transformations. Also, students should know that no cards will be used more than once and none will be left over.



Consider eliminating some of the cards for a shorter activity. Suggested cards for eliminating are card 5, card 6, card 9, and card 10. Consider giving patty paper or other manipulatives for students to use to engage kinesthetically with the physical transformations of the images.



Engage students in a discussion after the card sort before moving to the next component. As part of the discussion, ask students how they determined the solutions.



For a challenge, remove some of the written description and algebraic description cards and replace them with blank cards where the students must write the description that is missing for that set.

4.8.3 Wrap-Up: Reflection

Component Overview: Students will complete a self-reflection where they indicate where their current learning is related to the learning targets of this lesson. This activity should be done independently to allow for individual responses.



4.8.3 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is in relation to each target.

Answer will vary.

1. I can match the written descriptions of a sequence of transformations to the coordinate grid with the pre-image and image.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

2. I can match the algebraic descriptions of a sequence of transformations to the coordinate grid with the pre-image and image.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

3. I can determine a sequence of transformations that map one figure onto another congruent figure.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.



Refer to the Unit Guide for a digital, interactive extension on writing algebraic descriptions for transformations.